



Fall Semester 2025-26

EMPLOYEE DATABASE MANAGER PROJECT REPORT

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YOUR LEARNING DESTINATION



INTRODUCTION

Employee data management is essential for any organization. Traditional methods of maintaining records manually are inefficient and prone to errors. This project implements a Python-based console application that manages employee records using a CSV file. It covers file handling, modular programming, exception handling, and basic data analysis.





PROBLEM STATEMENT

Managing employee details manually results in inaccuracies and data loss. A digital solution is needed to maintain structured records, update salaries, and generate performance analytics.





FUNCTIONAL REQUIREMENTS

FR1: Employee Registration

The system must allow adding a new employee with details like:

Employee ID

Full Name

Designation

Salary

The system validates mandatory fields before saving.

FR2: Employee Data Storage

All employee information is stored in a CSV file.

The system ensures that no duplicate employee ID is stored.

FR3: Update Employee Information

The system allows updating the employee's Salary.

System validates updated values before saving.



FR4: Delete Employee Records

The system allows deletion of an employee record using Employee ID.

FR5: Display Employee List

The system displays all employee information.





NON-FUNCTIONAL REQUIREMENTS

NFR1: Performance

System saves, searches, updates, and deletes the employee data within 1 second.
Search operations returns results in real-time.

NFR2: Usability

User interface is simple, clear, and easy to understand.
Error messages are readable and informative.

NFR3: Reliability

System does not crash during data operations.
System maintains data integrity even after sudden program exit.

NFR4: Data Accuracy

No employee records are saved without mandatory details.
System prevents duplicate employee IDs.

NFR5: Portability

The program runs on Windows

NFR6: Maintainability

Comments and documentation are added wherever required.

NFR7: Scalability

System supports:
1000+ employee records





SYSTEM

ARCHITECTURE

1. User Interface Layer

Text-based menu for user interaction

Takes input and displays output

Calls appropriate functions

Component: `main()`

2. Application Logic Layer

Handles all operations:

Add employee

Read records

Update salary

Delete record

Calculate performance metrics

Performs validation and error handling

Components: CRUD functions + metrics function

3. Data Access Layer

Reads and writes data from/to CSV file

Converts CSV rows into dictionaries

Ensures proper file formatting

Components: `get_all_records()`, `write_all_records()`

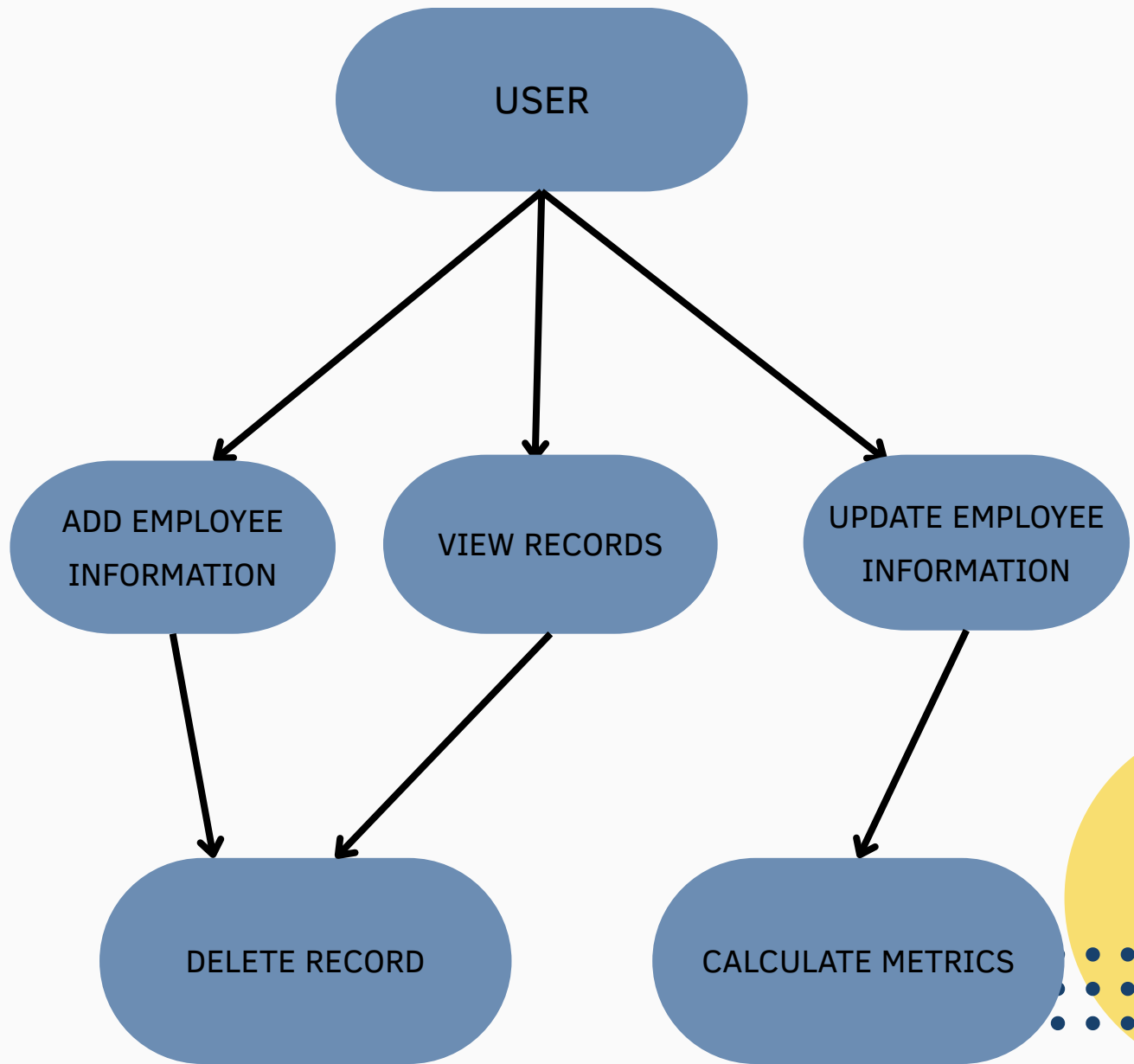
4. Storage Layer

Stores persistent employee data

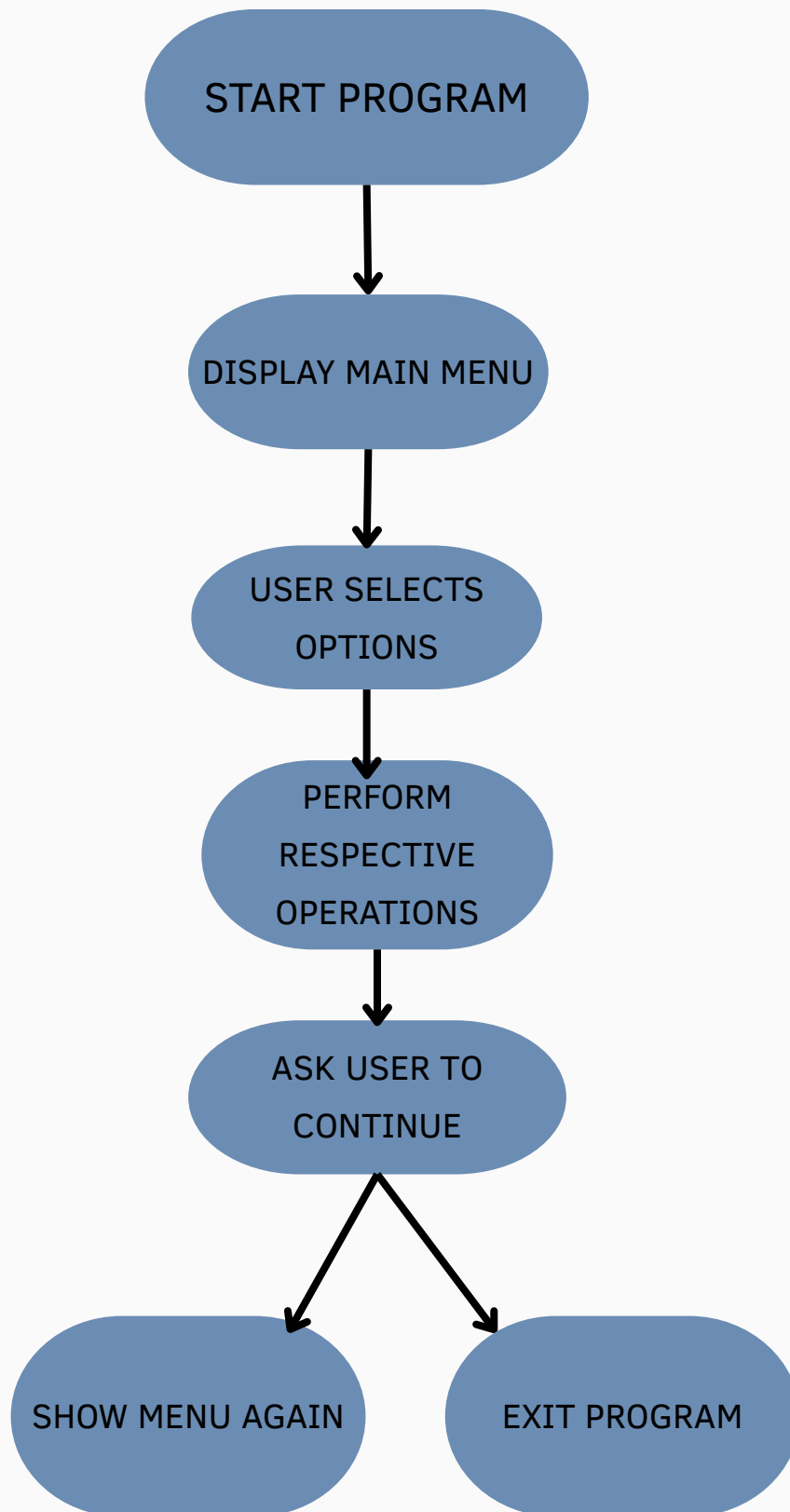
Component: `Employee.csv` (acts as a simple database)



USE CASE DIAGRAM

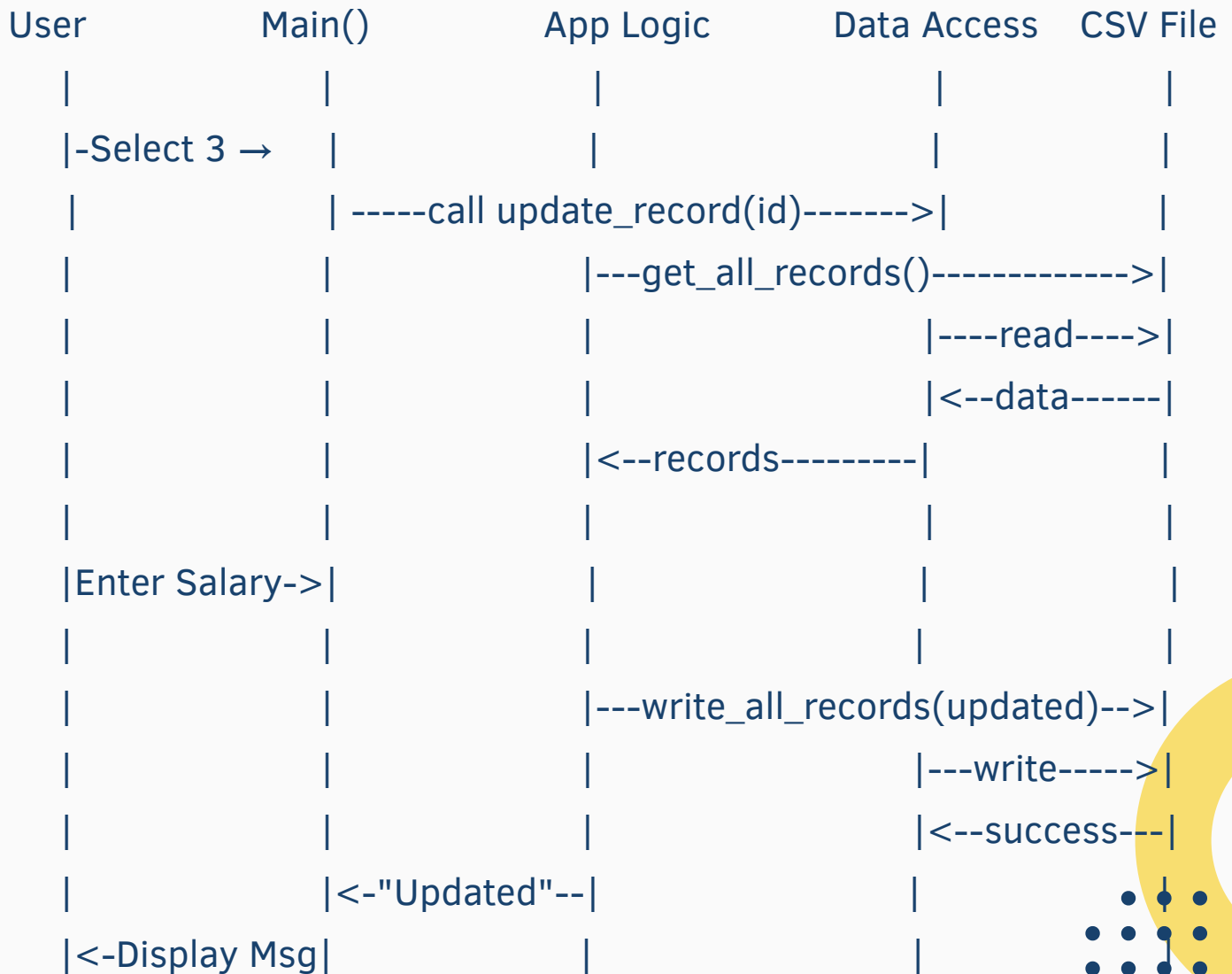


WORKFLOW DIAGRAM



SEQUENCE DIAGRAM

EXAMPLE: Updating Salary





DESIGN DECISIONS & RATIONALE

- CSV storage chosen for simplicity
- DictReader/Writer ensures structured data
- math module used for performance analysis
- Input validation prevents system errors
- Modular functions → easy to maintain



SCREENSHOTS / RESULTS

```
=====
Employee Database Manager (CSV)
=====
```

```
Enter 1 to write new records
Enter 2 to read all records
Enter 3 to update salary by ID
Enter 4 to delete record by ID
Enter 5 to calculate performance metrics (using math module)
Enter 6 to exit program
Enter choice:
```

Main Menu

```
--- Add New Records ---
```

```
Enter the number of records to add: 1
```

```
--- Record 1 ---
```

```
Enter Emp ID (numeric): 12001
Enter name: Abhishek Kumar
Enter salary (numeric): 2000000
Enter designation: Marketing Manager
```

```
Successfully entered and saved the new record(s)!!
```

**Add Employee
Screen**

```
=====
Employee Database Manager (CSV)
=====
```

```
Enter 1 to write new records
Enter 2 to read all records
Enter 3 to update salary by ID
Enter 4 to delete record by ID
Enter 5 to exit program
Enter choice: 2
```

```
--- Employee Records ---
```

Emp_ID	Name	Salary	Designation
12001	Abhishek Kumar	2000000	Marketing Manager
12304	Vineeta Sharma	3000000	Software Engineer
13023	Shweta Singh	2500000	MD

View Results

```
=====
Employee Database Manager (CSV)
=====
```

```
Enter 1 to write new records
Enter 2 to read all records
Enter 3 to update salary by ID
Enter 4 to delete record by ID
Enter 5 to exit program
Enter choice: 4
Enter the Employee ID to delete: 12001
```

```
--- Delete Record ---
```

```
Record with ID 12001 has been successfully deleted.
```

**Delete
Operations**



TESTING APPROACH

- Test cases include:
- Adding valid data
- Rejecting duplicate IDs
- Updating invalid ID
- Deleting valid/invalid IDs
- Salary validation
- CSV file integrity



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CHALLENGES FACED

- Handling numeric validation
- Maintaining clean CSV file
- Ensuring no data duplication
- Preventing crashes during user input





LEARNINGS & KEY TAKEAWAYS

- File handling in Python
- Practical use of csv and os modules
- Better control flow & error handling
- Real-world application development





FUTURE ENHANCEMENTS

- GUI interface (Tkinter)
- Login system (Admin/HR)
- Database integration (SQLite)
- Exporting analytics to charts



A decorative graphic consisting of a grid of dots in the top left corner, with some dots in grey and others in blue.

REFERENCES

- [Python Official Documentation](#)
- [W3Schools Python File Handling](#)
- [Instructor Notes](#)

